

Revolution Agriculture Maggot AI Ecosystem: Implementation of Artificial Intelligence to Maximize Agricultural Production by Utilizing Maggot as a Fertilizer Source

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Abstract. The management of organic waste in Indonesia is a growing concern due to its substantial amount and the harm it does to the environment. Using Black Soldier Fly (BSF) maggots as a decomposition agent to break down organic waste and use the resulting natural fertilizer is an efficient solution. In addition, by taking into account different soil parameters, artificial intelligence (AI) technology can assist in identifying the kind of plant that is best suited for a specific piece of land. In order to manage organic waste and boost agricultural productivity in a sustainable way, this research suggests a "Revolution Agriculture Maggot AI Ecosystem." By recommending crops based on environmental factors to prevent crop failure, the AI technology is anticipated to support food security.

INTRODUCTION

The main source of waste in Indonesia is currently organic waste, making waste management a complicated issue. Indonesia accumulates 23,733,245 tons of waste annually (KLHK, 2023). Food waste makes up 41.3% of the pile. But the annual decrease in waste—all types combined—is only about 16.4%. According to Puger, I.G.N. (2018), any remaining organic waste will decompose and release methane gas (CH₄) into the atmosphere, contributing to global warming and leaving behind an unpleasant stench.

Many creative solutions are beginning to be developed as a result of the massive increase in awareness of the urgency of organic waste management. Currently under development is the use of larvae of the Black Soldier Fly, or maggots (*Hermetia illucens*). The community can effectively deal with organic waste, which is a persistent issue, by using BSF Maggot to break it down (Goca et al., 2023).

The problem's context will get more complicated when it comes to issues with managing agricultural land. Farmers frequently struggle to choose the best kind of plant for a given plot of land (Sari FP et al. 2024). Artificial intelligence (AI) technology offers a significant remedy in this regard. Farmers will be able to choose crops more accurately and use maggot fertilizer to improve soil fertility by using artificial intelligence to thoroughly analyze a variety of soil factors and provide recommendations for the best crops.

The fact that climate change is having an effect on the agriculture industry highlights how urgent this problem is. According to Y Servina's 2019 study, climate change has negatively impacted tropical regions' ability to produce fruits and vegetables, necessitating the development of efficient agricultural adaptation techniques. As part of an adaptation strategy that promotes sustainable food security and aligns with Sustainable Development Goal (SDG) number 12, particularly with regard to responsible consumption and production, the use of maggots and AI-based organic waste management can be implemented.

Against this background, ideas for solutions were developed that included using maggots to reduce the accumulation of organic waste, using natural fertilizers to increase soil fertility and agricultural productivity, and using artificial intelligence to recommend plants based on the analysis of the characteristics of the land. The "Revolution Agriculture Maggot AI Ecosystem," as this solution is named, provides a comprehensive method for

managing organic waste and optimizing agricultural output. It is anticipated that the introduction of AI technology into this ecosystem will significantly lower greenhouse gas emissions and promote sustainable food security.

RESEARCH METHOD

The study is scheduled to take place in August and September of 2024. The National Institute of Technology, Bandung's Electronics Laboratory was the site of the prototype's design and production. The testing phase will take place in Lembang, Bandung's Cikole District. LED grow light 50 cm, waterproof DS18B20, DC fan, DC pump, PSU, ESP 32, panel box, regulator, nozzle, 4 channel relay, PCB, state button, emergency button, OLED LCD, DMS sensors, and NPK sensors are the materials used in this study.

There will be two stages to the research: modeling and testing. At the modeling stage, SolidWorks will be used to create a 3D design, Google Collabs will be used to model the Artificial Intelligence that will be used in this tool, and Proteus software will be used to create a series of tools that will be made in the form of a ground condition detection sensor based on Artificial Intelligence and also a maggot incubator

An Artificial Neural Network—a model that mimics how the human brain functions—is used in the process of developing an artificial intelligence model. A supervised machine learning algorithm is what ANN is, according to Dongare et al. (2012). A large number of basic processing units connected by threaded connections make up an ANN. Processing nodes are referred to as "neurons" in analysis. The information that is locally accessible at each node—either internally stored or obtained via weighted connections—determines that node's output alone. The ANN model is expressed mathematically as follows:

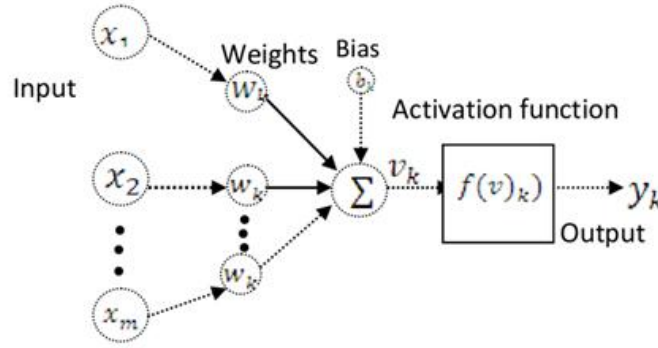


FIGURE 1. The ANN Mathematics Model

Based on this model, the neuron activity interval can be shown as

$$V_k = \sum_{j=1}^m w_{kj} x_j \quad (1)$$

The output of neuron Y_k is the result of several activation functions on V_k .

RESULT AND DISCUSSION

The system design is carried out using software and the following is the result of the design. According to FIGURE 2, the maggot incubator system is controlled by an ESP32 microcontroller. This microcontroller receives information from the DHT22 temperature and humidity sensor. The goal of using this tool is to maximize maggots as a decomposing agent for organic waste (Tomberlin et al., 2018). Tomberlin et al. (2002) found that the temperature of maggot growth media influences maggot production and growth rate. So it is critical to maintain stable temperature conditions. According to Irwan et al. (2022), BSF maggots can develop optimally at 32.2°C and 64.2% humidity, and monitoring of these conditions is required via incubator sensors.

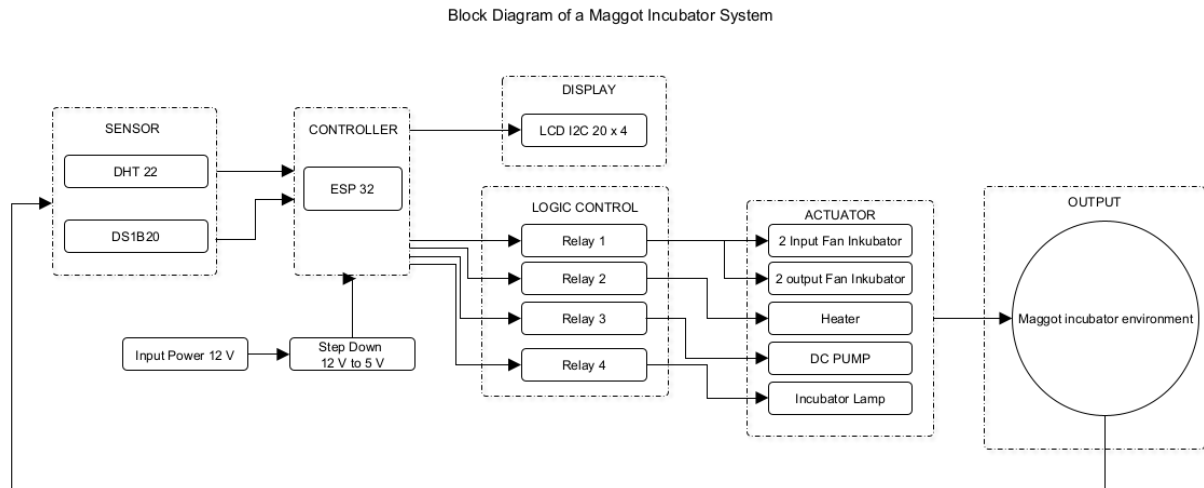


FIGURE 2. Block diagram of maggot incubator system

According to FIGURE 2, the microcontroller directs the relay module to regulate external devices such as heaters, lights, and water pumps in order to maintain optimal conditions in the incubator. The temperature and humidity monitoring results are shown on a 20 x 4 LCD. The step down module provides the system with the 5V voltage. All components work together to create the optimum habitat for maggot growth.

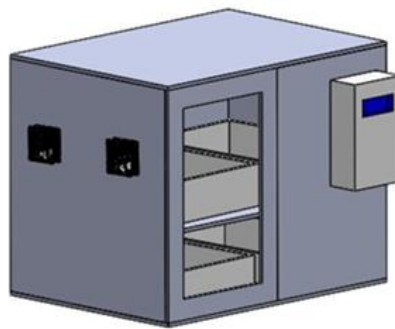


FIGURE 3. Design of the maggot incubator

FIGURE 3 depicts an incubator designed to provide optimal environmental conditions for maggots. This equipment includes sensors that measure environmental conditions in the incubator, such as a DHT22 sensor for room humidity and a DS18B20 temperature sensor for temperature monitoring. Actuators, such as air conditioning and heaters, help to adjust and balance environmental conditions. All systems operate in a closed loop that is controlled by the panel box, and temperature and humidity data can be displayed on the LCD screen located on the system panel box.

This maggot incubator includes two stages. The first stage, located in the bottom, is used to store eggs until they develop into baby maggots. After developing into a baby maggot with high environmental sensitivity, the maggot will be raised to the top to provide stable environmental conditions, hence aiding the maggot's growth and decomposition. This incubator also has a watering nozzle to keep the room humid and a growlight to promote maggot growth.

This maggot incubator allows maggots to be maximized in the process of digesting trash and preparing it for use as fertilizer. Based on data from Agustin et al's (2024) research, the technique will yield 2500 g cashgots and 561 g maggots in 14 days if a maximum of 200 g maggots are fed and mixed organic waste is used. This cashgot can yield 2% total NPK nutrients with a pH of 4.5-6.5, meeting 2019 Ministry of Agriculture criteria (Agustin et al., 2024). This makes it a viable alternative fertilizer for farmers.

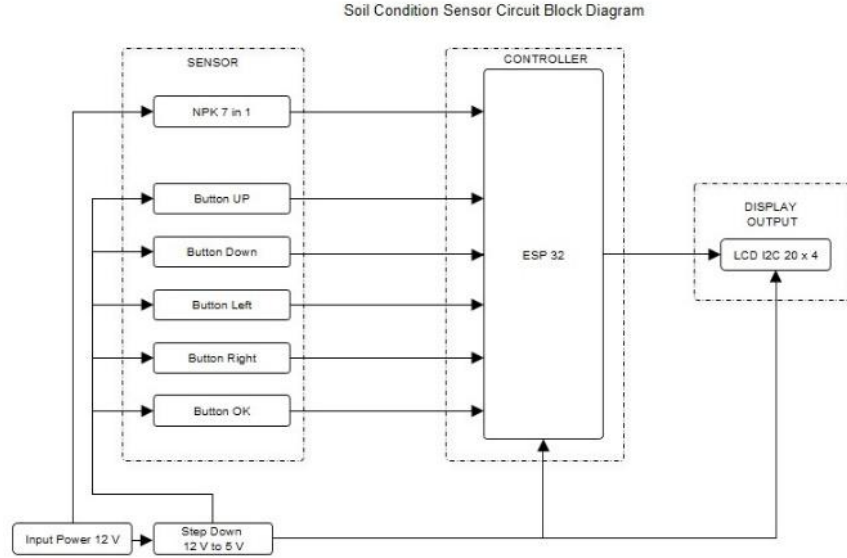


FIGURE 4. Soil Condition Sensor Block Diagram

The NPK sensor, shown at the FIGURE 4, detects nitrogen, phosphorous, and potassium levels in soil. This sensor transmits measured data to the microcontroller. Aside from that, temperature and humidity sensors measure the soil's thermal and moisture conditions and transfer this information to the microcontroller. This microcontroller is linked to a number of additional components, including an LCD screen that shows data from various sensors. There are also a few buttons or switches that allow for manual interaction with the system, as well as communication modules that can convey data to an artificial intelligence (AI)-powered platform. The AI system collects and analyzes sensor data to propose appropriate plant species depending on environmental circumstances. This technology aims to enhance crop choices by taking into account real-time soil conditions, resulting in more efficient and sustainable agriculture. design of the sensor device as shown in FIGURE 5, the soil condition sensor device is designed with a flexible form to accommodate various field applications

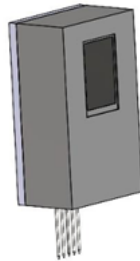


FIGURE 5. Soil Condition Sensor Device

Artificial intelligence experiments were done out with neural network models. In a neural network, each neuron takes the output from the neurons in the previous layer as input and passes it to the next layer. In ANN, the activation function is utilized to extract output from one layer and feed it into the next (Sharma & Athaiya, 2017). If the activation function is not employed in the neural network, the output signal will be a simple linear function, such as a first-degree polynomial. The activation functions employed during the procedure include ReLu, Leaky ReLu, and softmax. ReLu is a nonlinear function that is commonly utilized in ANNs. The advantage of ReLu is that not all neurons are stimulated simultaneously. The neuron will only be disabled if the linear transformation's output is zero. Ho, Y., and Wookey, S. (2019) define this function as:

$$f(X) = \max(0, x) \quad (2)$$

Aside from that, when developing this AI model, assessment metrics are employed to analyze the categorization model's performance. This report uses many equations to offer a detailed summary of how well the model performs on each class in the classification data.

The model architecture used in this AI modeling experiment is as follows.

TABLE 1. Model Architecture

Layer	Output shape	Param #
dense (Dense)	(none,128)	1,024
batch_normalization (BatchNormalization)	(none,128)	512
dropout (Dropout)	(none,128)	0
leaky_re_lu (LeakyReLu)	(none,128)	0
dense_1 (Dense)	(none,64)	8,256
dense_2 (Dense)	(none,32)	2,080
dense_3 (Dense)	(none,22)	7,26

The neural network model employed has numerous layers with the architecture shown in the table. First, there is a Dense layer with 128 neurons. This layer includes 1,024 parameters that reflect each neuron's weight and bias. This layer performs the major calculations by multiplying inputs by weights, accumulating biases, and applying activation functions. Furthermore, the model employs a Batch Normalization layer (batch_normalization), which similarly has an output shape of 128. This layer contains 512 parameters that are used to normalize the output of the preceding layer. The goal of batch normalization is to normalize the activations from the preceding layer, resulting in a more stable distribution that can speed up the training process and improve accuracy.

The Dropout layer prevents overfitting by randomly altering the fraction of neurons that are disregarded throughout the training process. Although this layer does not add parameters, it plays a vital role in increasing the model's generalization. Following that, there is a Leaky ReLU activation layer (leaky_re_lu), with an output of 128. This layer, with no extra parameters, plays a crucial role in addressing the vanishing gradient problem by giving tiny values (leaks) to dormant neurons. This model is subsequently followed by many more dense layers, including dense_1 with 64 neurons and 8,256 parameters, dense_2 with 32 neurons and 2,080 parameters, and dense_3 as the output layer with 22 neurons and 726 parameters. These thick layers are in charge of executing additional calculations that gradually abstract the input information to produce the final result.

Following training, this model had an accuracy of 0.9430 and a loss of 0.1377 on the training data. Using validation data, this model has a val_accuracy of 0.9773 and a val_loss of 0.0672. These FIGUREs demonstrate that the model performs admirably, with an overall accuracy of 0.9773, or almost 97.73%. This suggests that the model can predict quite well on data that it has never encountered before.

The following is a comparison of the loss during model training with the loss during testing at each epoch. FIGURE 6 shown graph compares training loss to validation/test loss while training a neural network model with several layers and a sophisticated design. The model includes several critical layers, including the initial Dense layer with 128 neurons and 1,024 parameters, followed by the Batch Normalization layer, which helps normalize the output with 512 parameters. In addition, a Dropout layer inhibits overfitting, and a Leaky ReLU activation layer solves the vanishing gradient problem. The graph illustrates that at the start of training, both training and validation losses are quite large, indicating that the model did not successful in learning patterns from the data. However, as the number of epochs rises, the training loss value lowers considerably, showing that the model is beginning to learn well. The validation loss value likewise reduces, reaching stability before the training loss, demonstrating that the model can generalize well to data not observed during training. The significance of layers such as Batch Normalization and Dropout is demonstrated by the relatively little difference between training loss and validation loss at the conclusion of training, showing that the model effectively avoided overfitting. Finally, our model obtained very high accuracy, namely 97.73% on validation data with a validation loss of 0.0672, demonstrating remarkable prediction performance.

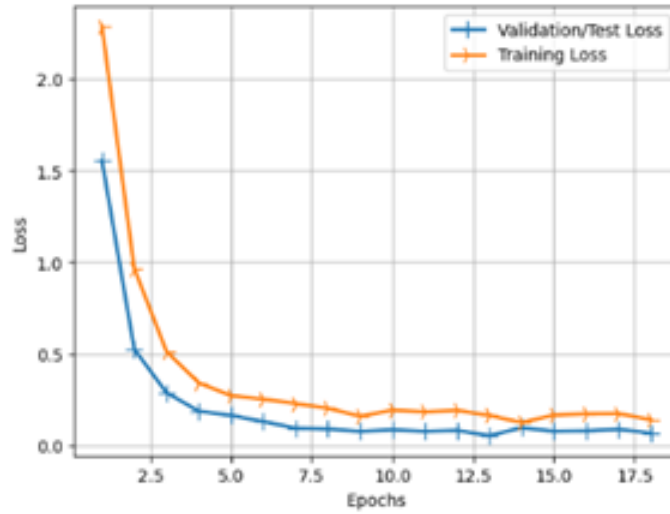


FIGURE 6. Comparison Graph of Training and Testing Loss

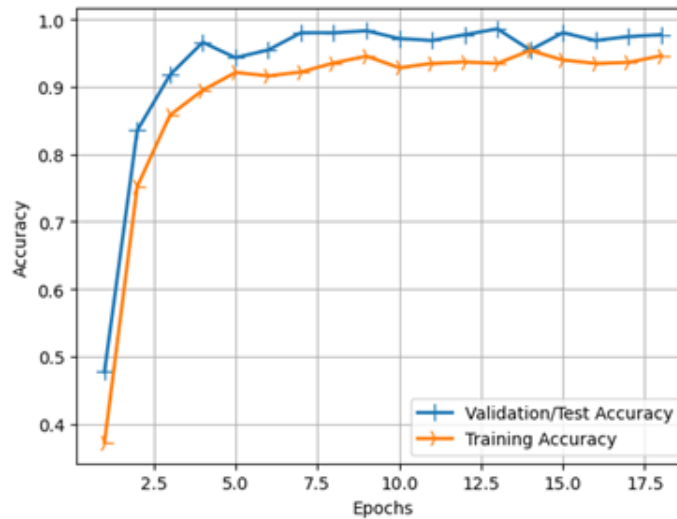


FIGURE 7. Comparison of Accuracy during Training and Testing

FIGURE 7 shown graph compares the training accuracy and validation/test accuracy of neural network models at each epoch. At the outset of training, both training and validation accuracy are very low, but they rapidly rise as the period progresses, indicating that the model learns fast from training data. After around five epochs, both accuracies begin to settle, with validation accuracy (shown in blue) tending to be greater and more stable than training accuracy (orange line). This shows that the model not only learns from the training data, but also can generalize well to data that was not encountered during training, indicating that it is not overfitting. At the end of training, both training and validation accuracy remained high and constant, suggesting that the model had reached peak performance in predicting with high accuracy on both training and validation data.

TABLE 2. Prediction Data with Dataset Test

True label	Input features							Top five perdition
	N	P	K	Temperature	Humidity	pH	Rainfall	
muskmelon	101.0	17.0	47.0	29.494	94.7298	6.18	26.308	muskmelon: 99.95% watermelon: 0.00% papaya: 0.00% mungbean: 0.00% rice: 0.00%
watermelon	98.0	8.0	51.0	26.179	86.522	6.25	49.430	watermelon: 99.68% muskmelon: 0.26% coffee: 0.00% maize: 0.00% cotton: 0.00%
Papaya	59.0	62.0	49.0	43.360	93.351	6.94	114.77	papaya: 99.94% jute: 0.01% apple: 0.00% rice: 0.00% banana: 0.00%
mango	18.0	19.0	27.0	27.775	52.346	4.77	94.112	mango: 99.46% mothbeans: 0.39% orange: 0.05% pomegranate: 0.02% kidneybeans: 0.01%

TABLE 2 shows the test results of an AI model that was trained on a plant dataset. The model predicts plant kinds based on a variety of input characteristics such as nitrogen (N), phosphorus (P), potassium (K), temperature (Temperature), humidity (Humidity), soil pH, and rainfall. Each row in the table displays the real label (original label) of the plant being tested, the value of the input feature, and the top five predictions made by the model, along with their confidence percentage. Given example, for a plant with a valid label of muskmelon, the model predicts with 99.95% confidence that the plant is a muskmelon based on inputs such as nitrogen (101.0) and soil pH (6.18), but other predictions such as watermelon and papaya have 0% likelihood, demonstrating confidence. The model scores highly on its primary predictions.

The model also performs well with plants labeled watermelon and papaya. The algorithm predicted that the watermelon plant was a watermelon with 99.68% confidence, and while muskmelon had a minor chance (0.26%), the other predictions were almost impossible. Similarly, for papaya plants, the model is nearly certain (99.94%) that the plant is papaya, with extremely low probability for other possibilities such as jute (0.01%). The AI algorithm accurately and confidently predicts plant kinds based on input characteristics. Apart from using the dataset, testing is carried out with random data for model testing.

TABLE 3. Prediction Data with Random Test

Random case -	Input features							Top five prediction
	N	P	K	Temperature	Humidity	pH	Rainfall	
1	54.0	45.5	46.7	39.89	76.66	8.72	129.88	papaya: 89.64% jute: 6.18% apple: 1.77% pigeonpeas: 0.66% rice: 0.45%
2	105.1	32.5	202.4	40.997	74.264	6.5	129.021	Top 5 Predictions with probabilities: coffee: 85.06% watermelon: 13.21% chickpea: 1.54% mango: 0.05% apple: 0.04%
3	128.0	126.7	116.3	17.068	70.37	7.5	125.79	banana: 99.95% chickpea: 0.00% apple: 0.00% jute: 0.00% maize: 0.00%
4	78.4	91.2	9.63	16.259	72.742	5.4	25.822	maize: 63.23% mothbeans: 36.72% blackgram: 0.00% pigeonpeas: 0.00% pomegranate: 0.00%

The AI model experiment with four random test cases predicts plant kinds based on input characteristics such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, soil pH, and rainfall. In Random Test Case 1, using input characteristics such as N (54.06), P (45.57), and temperature (39.89°C), the model predicts papaya plants with the highest likelihood of 89.64%, followed by jute (6.18%), and various other options with lesser probabilities. In Random Test Case 2, if the input features include N (105.10), K (202.45), and temperature (40.99°C), the model predicts coffee plants with an 85.06% probability, followed by watermelon with a probability of 13.21%, demonstrating that these features are more closely associated to the coffee plant.

Furthermore, in Random Test Case 3, with input characteristics such as N (128.07), P (126.73), and a lower temperature (17.07°C), the algorithm is almost totally certain (99.95%) that the plant is banana, but other predictions such as chickpea and apple have a almost 0% chance. Finally, in Random Test Case 4, the model forecasts maize as the most likely crop (63.23%), based on inputs such as N (78.47), P (91.24), and temperature (16.26°C), followed by mothbeans (36.72%). This data demonstrates how AI models may use a variety of agronomic variables to forecast crop varieties with varied levels of accuracy depending on the input circumstances.

CONCLUSION

This study found that combining the use of maggots as organic waste decomposers with artificial intelligence technology may considerably boost agricultural output and soil fertility. Implementing the "Revolution Agriculture Maggot AI Ecosystem" technology reduces greenhouse gas emissions while simultaneously promoting long-term food security. The AI model, trained on various agronomic parameters, exhibited high accuracy (up to 97.73%) in predicting suitable crop types based on real-time soil and environmental data. This confirms the system's potential to optimize land use, reduce crop failure risks, and promote efficient, eco-friendly agricultural practices in the face of climate change.

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